

ECON 3350/7350 Applied Econometrics for Macroeconomics and Finance

Lecture 8: Volatility Models (2)

Fu Ouyang

School of Economics
University of Queensland

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Volatility Modeling Extensions

Recommended readings

Author	Title	Chapter	Call No
Enders	Applied Econometric Time Series, 4e	3	HB139 .E55 2015
Brooks	Introductory Econometrics for Finance, 3e	9	HG173 .B76 2014
Verbeek	A Guide to Modern Econometrics	8.10	HB139 .V465 2012

ARCH(q) and GARCH(p, q)

ARCH(q) process:

$$\begin{aligned} E(\varepsilon_t^2 \mid \varepsilon_{t-1}^2, \dots, \varepsilon_{t-q}^2) &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \dots + \alpha_q \varepsilon_{t-q}^2, \\ &= \alpha_0 + \alpha(L) \varepsilon_{t-1}^2, \end{aligned}$$

where $\alpha(L)$ is a lag polynomial of order $q - 1$, $\alpha(1) < 1$, $\alpha_j > 0$ for $j = 0, \dots, q$.

GARCH(p, q) process:

$$\begin{aligned} h_t &= \alpha_0 + \sum_{j=1}^q \alpha_j \varepsilon_{t-j}^2 + \sum_{j=1}^p \beta_j h_{t-j}, \\ &= \alpha_0 + \alpha(L) \varepsilon_{t-1}^2 + \beta(L) h_{t-1}. \end{aligned}$$

In practice, focus is on GARCH(1, 1), where non-negative h_t is obtained by restricting $\alpha_0 \geq 0$, $\alpha_1 \geq 0$, and $\beta_1 \geq 0$.

Regressions and Autoregressions

Consider a **regression** for the mean of y_t with ARCH(q) errors:

$$y_t = \beta_1 + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t,$$
$$\varepsilon_t = \nu_t \sqrt{\alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2},$$

where $\nu_t \sim \mathcal{N}(0, 1)$ and $E(\varepsilon_t \nu_{t-s}) = 0$ for all t, s .

Consider an **auto-regression** for the mean of y_t with GARCH(1, 1) errors:

$$y_t = a_0 + a_1 y_{t-1} + \varepsilon_t,$$
$$\varepsilon_t = \nu_t \sqrt{h_t},$$
$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1},$$

where $\nu_t \sim \mathcal{N}(0, 1)$ and $E(\varepsilon_t \nu_{t-s}) = 0$ for all t, s .

Extensions to the Basic GARCH Model

Huge number of extensions and variants of the GARCH model proposed in the past three decades:

- **practical** problems with basic GARCH: **non-negativity constraints** difficult to impose;
- **conceptual** limitations in GARCH: cannot account for **leverage effects**.

Some of the most **important extensions** are:

- Asymmetric models: TGARCH and EGARCH (account for leverage effects);
- (G)ARCH-M models: particularly suited to study asset markets;
- IGARCH models: impose constraints that account for the persistence of volatility.

The EGARCH Model

EGARCH (suggested by Nelson, 1991) specifies the variance equation as:

$$\ln h_t = \alpha_0 + \beta_1 \ln h_{t-1} + \lambda \frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}} + \alpha_1 \left| \frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}} \right|.$$

Advantage of EGARCH is h_t will be positive for all values of parameters (due to the specification in logs).

$\nu_{t-1} = \frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}}$ is iid **standard normal**; $|\nu_{t-1}| = \left| \frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}} \right|$ is iid **half-normal**
with $E(|\nu_{t-1}|) = \sqrt{2/\pi}$.

Leverage Effect in EGARCH

The effect of a shock in ε_t on h_t :

- if $\frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}}$ is **positive**, the effect on $\ln h_t$ is $\alpha_1 + \lambda$;
- if $\frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}}$ is **negative**, the effect on $\ln h_t$ is $\alpha_1 - \lambda$.

Effect of a negative shocks is **greater** than the effect of a positive shock *iff*

$$\begin{array}{ccc} \text{positive shock} & & \text{negative shock} \\ \underbrace{\alpha_1 + \lambda} & < & \underbrace{\alpha_1 - \lambda} \quad , \\ \lambda < 0. \end{array}$$

This is the **leverage effect**: volatility increases more when stock prices fall than when they rise; if $\lambda = 0$ then no leverage effect exists and if $\lambda > 0$ then a reverse effect exists.

The Leverage Effect

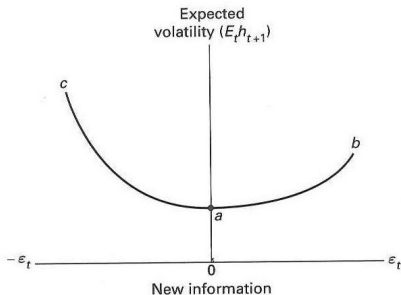


FIGURE 3.11 The Leverage Effect

The standardized $\frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}}$ is unit free and permits a more natural interpretation of the size and persistence of the shocks.

Source: Enders (2015)

Volatility Prediction–EGARCH

Assuming the parameters are known, the natural predictor is

$$h_t = \exp(\ln h_t) = \exp(\alpha_0 + \beta_1 \ln h_{t-1} + \lambda \nu_{t-1} + \alpha_1 |\nu_{t-1}|),$$
$$E_t(h_{t+j}) = E_t(\exp(\ln h_{t+j})), \quad j > 0.$$

When in addition shocks are Gaussian,

$$\ln h_t = (1 - \beta_1)\alpha_0 + \beta_1 \ln h_{t-1} + g(\nu_{t-1}),$$
$$g(\nu_{t-1}) = \lambda \nu_{t-1} + \alpha_1 \left(|\nu_{t-1}| - \sqrt{2/\pi} \right).$$

The one-step ahead forecast is given by

$$h_{t+1} = h_t^{\beta_1} \exp((1 - \beta_1)\alpha_0) \exp(g(\nu_t)).$$

The TGARCH or GJR Model

The **threshold** GARCH (TGARCH) model, developed by Glosten, Jaganathan and Runkle (1994), is

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \lambda d_{t-1} \varepsilon_{t-1}^2 + \beta_1 h_{t-1},$$
$$d_{t-1} = \begin{cases} 1 & \text{if } \varepsilon_{t-1} < 0, \\ 0 & \text{otherwise.} \end{cases}$$

We require $\alpha_0 \geq 0$ and $\alpha_1 \geq 0$ for non-negativity.

The effect of an ε_{t-1} shock on h_t :

- if $\varepsilon_{t-1} \geq 0$, $d_{t-1} = 0$ and the effect is $\alpha_1 \varepsilon_{t-1}^2$;
- if $\varepsilon_{t-1} < 0$, $d_{t-1} = 1$ and the effect is $(\alpha_1 + \lambda) \varepsilon_{t-1}^2$;
- interpret as **leverage effect** when $\lambda > 0$.

Testing for Leverage Effects with GARCH

Estimate a GARCH and store the **standardized** residuals

$$\hat{\nu}_t = \frac{\hat{\varepsilon}_t}{\sqrt{\hat{h}_t}}, \quad \text{for } t = 1, \dots, T.$$

Use an F -test from the regression:

$$\hat{\nu}_t^2 = a_0 + a_1 \hat{\nu}_{t-1} + a_2 \hat{\nu}_{t-2} + \dots + u_t.$$

Under the null $H_0 : \alpha_1 = \alpha_2 = \dots$, the standardized square errors are uncorrelated with history of levels, and therefore, no leverage effects exist. Under H_1 , there are some leverage effects present. The usual F -statistic and critical values can be used.

Alternatively, use a t -test in the regression:

$$\hat{\nu}_t^2 = a_0 + a_1 d_{t-1} + u_t,$$

where $H_0 : a_1 = 0$ indicates no leverage effect and the usual t -statistic and critical values are applicable.

Testing for Leverage Effects with EGARCH or TGARCH

Recall that in both the EGARCH and TGARCH, leverage effects are controlled by a single parameter λ .

If either model is estimated, only a simple **one-tailed** t -test on λ is needed to test for leverage effects:

$$H_0 : \lambda = 0,$$

$$H_1 : \lambda < 0 \text{ for the EGARCH or } \lambda > 0 \text{ for the TGARCH.}$$

TGARCH Example

Data: monthly S&P 500 returns, 1979M12–1998M6. Source: Brooks (2002).

Estimated model is

$$\hat{y}_t = 0.172,$$

$$(3.198)$$

$$h_t = 1.243 + 0.015\varepsilon_{t-1}^2 + 0.604d_{t-1}\varepsilon_{t-1}^2 + 0.498h_{t-1}.$$

$$(16.37)$$

$$(0.44)$$

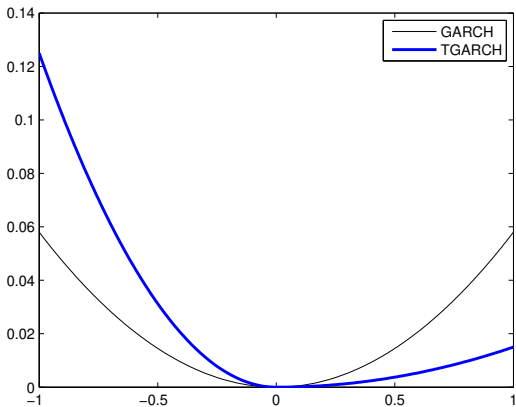
$$(5.77)$$

$$(14.99)$$

Test for leverage effects with $H_0 : \lambda = 0$ against $H_1 : \lambda > 0$; t -stat is larger than 1.645, so conclude **significant leverage effects**.

TGARCH Example

The **news impact curve** plots the next period volatility (h_{t+1}) that would arise from various positive and negative values of ε_t^2 .



(G)ARCH-in Mean

We expect a risk to be compensated by a higher return; so why not let the return be partly determined by risk?

Engle, Lilien and Robins (1987) suggested the ARCH-M specification:

$$y_t = \mu_t + \varepsilon_t,$$
$$\mu_t = \beta + \delta \sqrt{h_t} = \beta + \delta \sqrt{\alpha_0 + \alpha_1 \sum_{j=1}^q w_j \varepsilon_{t-j}^2},$$

where $\alpha_0, \alpha_1, \beta$ and $\delta > 0$ are constants; $w_1, \dots, w_q$ are **weights** assigned to past q squared errors (ELR set $w_j = (5 - j)/10$ for $j = 1, \dots, 4$).

(G)ARCH-in Mean

In ELR,

- y_t : excess return from holding a long-term asset relative to a one-period treasury bill;
- μ_t : risk premium necessary to induce the risk-averse agent to hold the long-term asset rather than the one-period bond;
- ε_t : unforecastable shock to the excess return on the long-term asset.

The ELR ARCH-M Model

The **expected excess return** from holding the **long-term asset** must be just equal to the **risk premium**:

$$E_{t-1}y_t = \mu_t.$$

The assumption is that the **risk premium** is an **increasing function** of the **conditional variance** of ε_t^2 .

- The greater the conditional variance of returns, the greater the compensation necessary to induce the agent to hold the long-term asset.

ELR Data

ELR use excess yields on six-month treasury bills:

- r_t : quarterly yield on a three-month treasury bill held from t to $t + 1$;
 - rolling over all proceeds; at the end of two quarters, an individual investing \$1 at the beginning of t will have $(1 + r)(1 + r_{t+1})$ dollars
- R_t : quarterly yield on a six-month treasury bill, buying and holding the six-month bill for the full two quarters will result in $(1 + R_t)^2$ dollars.

The **excess yield** y_t due to holding the six-month bill is

$$y_t = (1 + R_t)^2 - (1 + r_{t+1})(1 + r_t),$$

which is approximately equal to

$$y_t = 2R_t - r_{t+1} - r_t.$$

ELR ARCH Test

Estimates of a **constant mean** and **constant variance** model:

$$\hat{y}_t = 0.142 + \varepsilon_t.$$

(4.04)

The excess yield of 0.142% per quarter is more than **four standard deviations** from zero.

$LM_{ARCH} = 10.1$, which is larger than the critical value of 6.635 obtained from χ^2_1 at the 1% significance level.

The post-1979 period showed higher volatility than the earlier period.

ELR ARCH-M Estimates

The ARCH-M estimates are:

$$\hat{y}_t = -0.0241 + 0.687\sqrt{\hat{h}_t},$$

(-1.29) (5.15)

$$\hat{h}_t = 0.0023 + 1.64 (0.4\varepsilon_{t-1}^2 + 0.3\varepsilon_{t-2}^2 + 0.2\varepsilon_{t-3}^2 + 0.1\varepsilon_{t-4}^2).$$

(1.08) (6.30)

Results:

- Risk premium is time-varying.
- Estimate of 1.64 implies the **unconditional variance** is **infinite** (although conditional variance is finite).
- During volatile periods, the risk premium rises as risk-averse agents seek assets that are conditionally less risky.

Stochastic Volatility Models

An alternative to GARCH are the **stochastic volatility** models (SV).

The foundation of SV models differs from GARCH in that conditional variance is **not a deterministic function** of past shocks.

SV models are particularly designed to deal with **volatility clustering**.

SV models contain a **second error term** that enters the conditional variance specification; they are commonly written as **state-space** models.

Stochastic Volatility Models

A typical SV model is defined as

$$\begin{aligned}y_t &= a_0 + e^{\frac{h_t}{2}} \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, 1), \\ \beta(L)h_t &= \alpha_0 + \eta_t, & \eta_t &\sim \mathcal{N}(0, \sigma_\eta^2).\end{aligned}$$

$\beta(L)$ is a lag polynomial of order m ; the parameters in the model are $a_0, \alpha_0, \beta_1, \dots, \beta_m$.

Parameters of the SV model can be estimated using **maximum likelihood** or **simulation methods**.

SV Example

A SV model for the daily returns of an asset price y_t can be specified as

$$\begin{aligned}y_t &= a_0 + e^{\frac{h_t}{2}} \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, 1), \\h_t &= \phi h_{t-1} + \eta_t, & \eta_t &\sim \mathcal{N}(0, \sigma_\eta^2),\end{aligned}$$

and $0 < \phi < 1$.

The **signal** or **state**, h_t , is the **logarithmic volatility**, which we wish to estimate.

Volatility is mapped to **observed daily returns**, and it is assumed to be a **stationary** AR(1) process in this case.

Realized Volatility

The conditional variance is **latent** (i.e. not observed).

GARCH and SV type models produce **estimates** of the latent **conditional variance**.

Latent conditional variance models do not capture **high excess kurtosis** of returns—i.e., low but persistent autocorrelations in **squared returns**.

Merton (1980): we can accurately estimate the variance over a fixed interval with sufficiently **high frequency** data.

Realized Volatility

Andersen and Bollerslev (1998): estimate ex-post daily foreign exchange volatility by **sum of squared returns**, sampled at **5-minute intervals**.

The 5-minute frequency is a tradeoff between **accuracy** and **microstructure noise**.

Ex-post volatility measured this can be regarded as **observable**; it can be taken as **data** and modeled directly.

Further reading:

McAleer, M. and Medeiros, M. C. (2008), “Realized Volatility: A Review”, *Econometric Reviews*, 27(1), pp. 10–45.